AD - 303 (GS)

B.Tech 111 semester

Examination, December 2022

Grading System (GS)

Data Structure

Time : 3 Hours Maximum Marks :70

Note :i) Attempt any five questions .

ii) All questions carry equal marks .

1a)Explain Red Black Trees , also discuss the properties of Red Black Tree .

1b) Sort using quick sort algorithm –
36 , 25 , 32 , 5 , 8 , 65 , 38 , 47 , 95 .

2a) What is Hashing ? Explain different Hash function in detail .

2b) Explain AVL Tree . Insert the following elements in AVL Search Tree .

48 25 10 5 7 3 28 20 8 16

3 .a) Explain analysis of Heap Operation with example .

3b) Sort the following list using quick sort .

44 , 33 , 11 , 55 , 77 , 90 , 40 , 60 , 99 , 22 , 88 , 66 .

4 . a) Write the algorithm of insertion sort and find its Time complexity .

b) Difference Between Internal sorting and External sorting .

5 . a) Give the solution for following Recursion – $T(n) = 2T[n/2] + n \log n$.

b) Difference between Breadth First Search and Depth First Search

6 . a) Write an algorithm for insertion and deletion in priority Queues .

b) Explain Dijkastra's algorithm for finding shortest path with an exampl

7 . a) write short note on Internal trees . Explain with example .

b) How can you covert an infix expression to postfix operation using stac Give one Example .

8 . a) Write an algorithm Which counts the number of connected component a graph .

b) Construct a Tree for the given Inorder And Postorder traversals .

Inorder : DGBAHEICF

Postorder: GDBHIEFCA